

Abhisek Ghosal

Department of Chemistry, Northwestern University
2145 Sheridan Road, Evanston, IL 60208-3113
Phone: (+1) 708-971-4709 (+91) 954-784-2600
Date of Birth: 30th Nov. 1990
Email: · abhisek.ghoshal1990@gmail.com · abhisek.ghosal@northwestern.edu
Web: · abhisek.github.io



Research Experience

Northwestern University, Department of Chemistry

Evanston, IL, USA

Postdoctoral Research Scholar, Prof. George C. Schatz Group

Sep. 2023-Present

- **Theoretical characterization of autoionizing excited states and their role in chemical dynamics.**
 - Developing non-Hermitian Hamiltonian based quantum chemical methods for Feshbach and shape-type of resonances.
 - Computing the rate of collision-induced associative ionization processes in plasma formation.
 - Modelling the photodynamics of Green Fluorescent Protein involving autoionizing excited-state surfaces.
- **Electron–electron scattering–induced relaxation (EESR) of plasmonic hot carriers (HCs) from a many-body theoretical framework and its role in photocatalysis.**
 - Developing an efficient implementation of the self-energy at the GW/RPA level using multipole expansion for large plasmonic systems.
 - Modeling tractable simulations of quantum plasmonic processes relevant to photocatalysis, including EESR of plasmonic HCs.

TIFR Mumbai, Department of Chemical Sciences

Mumbai, MH, India

Visiting postdoctoral fellow, Dr. Vamsee K. Voora Group

May. 2021-Aug. 2023

- **Theoretical characterization of negative-ion resonances using complex density functional theory**
 - Developing efficient one-particle nonlocal exchange–correlation potentials by combining exact-exchange with random phase approximation (RPA) correlation.
 - Accessing negative-ion resonances by perturbing real, Hermitian nonlocal exchange–correlation (XC) potentials with complex absorbing local potentials.
- **Modeling the optical gap of F-center defects using many-body theory**
 - Developing efficient one-particle nonlocal exchange–correlation (XC) potentials by combining exact exchange with single-pole RPA correlation.
 - Accessing bulk and surface F-center defects by combining RPA based potentials with periodic electrostatic embedding.

IISER Kolkata, Department of Chemical Sciences

Mohanpur, WB, India

Graduate Researcher, Prof. Amlan K. Roy Group

Oct. 2014-Apr. 2021

- **Development of real-space density functional theory (DFT) for many-electron**

systems: electronic structure and dynamics

- Development of DFT methods within a Cartesian coordinate grid (CCG) framework.
- Methodological development of efficient Hartree–Fock exchange energy densities for hyper exchange–correlation (XC) functionals.
- Self-consistent and systematic optimization of range-separated hybrid functionals from *first principles*.
- Real-time time-dependent DFT in CCG for investigating nonlinear optical properties of many-electron systems.
- Modelling Charge-transfer excited states through Becke’s exciton model within a Hybrid generalised Kohn-Sham framework.

Education

Indian Institute of Science Education and Research, Kolkata

PhD., Theoretical Chemistry

Mohanpur, WB

2014 - 2021

- CSIR/UGC-SRF Awardee Dec-2016
- CSIR/UGC-JRF Awardee Dec-2014
- **Research Advisor:** Prof. Amlan K Roy

Presidency University

M.Sc., Physical Chemistry

Kolkata

2011-2013

- Degree Obtained June 2014
- **M.Sc. Thesis Advisor:**
Prof. Sanjib Ghosh, former Vice-Chancellor of Presidency college

Presidency College, University of Calcutta

B.Sc., Chemistry(Honours), Physics and Mathematics

Kolkata

2009-2011

Bhanderhati B.M. Institution, WBSHSE

High School graduated,

Hooghly

2009

Fellowships/Grants/Awards/Honors

- **Early Career Researcher Prizes, Molecular Physics** July 2022
(*An International Journal at the Interface Between Chemistry and Physics*)
- **International Travel Grant, DST-SERB, Gov. of India** Aug. 2019
- **Senior Research Fellowship, CSIR/UGC SRF, Gov. of India** Dec. 2017
- **Junior Research Fellowship, CSIR/UGC JRF, Gov. of India** Dec. 2014
- **Best Poster, Departmental Day of Chemical Sciences, IISER Kolkata** Dec. 2015
- **Secured a Leading Position, JAM-IITs, Gov. of India** 2011
(*Joint admission in Masters - Indian Institute of Technologies*)

Teaching Experience

IISER Kolkata, Department of Chemical Sciences

Mohanpur, WB

- CH4102: Molecular Structure and Symmetry

IISER Kolkata, Department of Chemical Sciences

Mohanpur, WB

Teaching Assistant

Autumn Semester, 2015

- CH4102: Molecular Structure and Symmetry

Proposed Teaching Contributions

At undergraduate level

- Classical Thermodynamics
 - Systems, properties, changes and processes, Heat and Work.
 - Energy and First law of Thermodynamics, Enthalpy and applications.
 - Exact Differential, Maxwell's Relations and Thermodynamic Temperature scale.
 - Second law of thermodynamics, Concept of Entropy and change in processes.
 - Equilibrium Constant, Le Chatelier principle, Chemical affinity and Free energy.
 - Chemical Potential, Conductance, Adsorption and surface tension.
- Statistical Thermodynamics
 - Microstate and thermodynamic probability, Statistical concept of entropy.
 - Boltzmann distribution formula, Partition function.
 - Fermi-Dirac and Bose-Einstein statistics.
- Basic Quantum Mechanics
 - Black-body radiation, Compton effect, Wave-particle duality, Uncertainty principle.
 - Operators and observables, Schrödinger wave equation, Particle in a 1D-3D box.
 - Simple Harmonic Oscillator, Two-particle rigid rotor, Hydrogen atom.
- Symmetry and Molecular Spectroscopy
 - Representation of symmetry, Great orthogonality and related theorems.
 - Molecular Hamiltonian, Potential energy surfaces and Group theoretical representation.
 - Rotational and Vibrational spectra, Vibrational spectra of polyatomic molecule.
 - Fluorescence, Phosphorescence and Predissociation spectra.
- Gas, Liquid and Solid State Physics
 - Kinetic theory of gases, Real gas and concept of Intermolecular Forces.
 - Characterization of Liquid state, Surface tension, Viscosity and colloid.
 - Macroscopic theory of dielectric dispersion, Debye and Onsager equations.
 - Characterization of Solid states, Unit cell, crystal system and Crystal planes.
 - X-ray diffraction, Intensity and structure determination, Techniques of crystal analysis.
 - Classical and Quantum theory of Heat capacity.
- Chemical Kinetics
 - Rate of reaction, Order of the reaction, Theory of reaction rate and Complex reactions.
 - Catalysis, Auto-catalytic reaction and Kinetic salt effects.
 - Photochemistry, Photosensitized Reaction and determination of quantum yield.
- Basic Mathematics in Chemistry
 - Differential calculus of one variable, Integral calculus and calculus of many variables.

- Infinite series, Complex numbers, functions of a complex variable and vector calculus.
- Matrices and Determinants, Linear vector spaces and Differential equations.

At advanced (Int-PhD/Ph-D) level

- Advanced Quantum Mechanics
 - *Principles of Quantum Mechanics*, by Prof. R. Shankar.
 - *Quantum Mechanics, Vol-1,2 and 3*, by Prof. C. Cohen-Tannoudji.
 - *Introduction to Quantum Mechanics in Chemistry*, by Prof. G. C. Schatz and Prof. M. A. Ratner.
- Advanced Electronic Structure
 - *Modern Quantum Chemistry: Introduction to Advanced Electronic Structure Theory*, by Prof. A. Szabo and Prof. N. S. Ostlund.
 - *Molecular Electronic-Structure Theory*, by Prof. T. Helgaker, Prof. P. Jorgensen and Prof. Jeppe Olsen.
- Modern DFT and TDDFT
 - *Density-Functional Theory of Atoms and Molecules*, by Prof. R. G. Parr and Prof. W. Yang.
 - *Density Functional Theory*, by Prof. E. Engel and Prof. R. M. Dreizler.
- Group Theory and Symmetry
 - *Group theory and Chemistry*, by Prof. D. M. Bishop.
 - *Molecular symmetry and Group theory*, by Prof. R. L. Carter.
- Integrated ML and AI approach in Chemistry
 - Basic Python programming and introduction to computational chemistry.
 - Introduction to ML model for example Scikit-learn and data structure.
 - Training and test data set and application to material sciences.
 - Chemical descriptor and Large-Language based AI model for data bank.

Mentoring Experience

- **Mr. Kishalay Mahato** BS-MS 2014, IISER Kolkata
(*University of Maryland, USA*)
- **Mr. Tarun Gupta** BS-MS 2015, IISER Kolkata
(*University of Warsaw, Poland*)
- **Mr. Raj Roy** JRF 2020, IISER Kolkata
(*New York University, Shanghai*)
- **Mr. Sayan Adhikari** BS-MS 2021, IISER Berhampur
(*University of California, Merced*)

Presentations

- **Symposiums**

- Abhisek Ghosal and George C. Schatz, CMQT Symposium, Northwestern University, (May 1-2, 2025).
- Abhisek Ghosal and George C. Schatz, IIN symposium, Northwestern University, (October 10, 2024).

- **Invited Talks**

- *Complex Density Functional Theory : A Non-Hermitian approach to Negative ion Resonances*, CASUS Institute Seminar, Helmholtz-Zentrum Dresden-Rossendorf e.V. (HZDR), October 8, (2024).
- *Theoretical basis and accurate approximations for modelling negative ion resonances*, Prof. William Glover Group, NYU Shanghai, June 26, (2024).
- *Taming negative ion resonances using non-local density functionals*, Prof. Stefano Sanvito Group, Trinity College, Dublin, November 17, (2023).
- *Density functional theory of many-electron systems in Cartesian grid*, on GIAN course of MHRD, Gov. of India organized by IIT-Bhubaneswar. (8-18 July, 2016).

- **Posters**

- Abhisek Ghosal and Amlan K Roy, *A self-consistent systematic optimization of range-separated hybrid functionals from first principles*, Virtual conference on recent development of electronic structure method, Department of Physics, UC Merced. (2-4 June, 2020).
- Abhisek Ghosal and Amlan K Roy, *A self-consistent systematic optimization of range-separated hybrid functionals from first principles*, 2020 New horizons in density functional theory Faraday discussions, Royal Society of Chemistry, UK. (2-4 September, 2020).
- Abhisek Ghosal and Amlan K Roy, *Real-time time-dependent density functional theory of many-electron systems in Cartesian grid*, 2019 TDDFT School & Workshop: Excited states and dynamics, Rutgers University-Newark. (5-14 August, 2019).
- Abhisek Ghosal and Amlan K Roy, *A pseudopotential Cartesian grid based KS-DFT for many-electron systems*, on Theoretical Chemistry symposium organized by School of Chemistry, University of Hyderabad. (14-17 Dec. 2016).
- Abhisek Ghosal and Amlan K Roy, *DFT of many electron systems in Cartesian grid*, on Departmental Day of Chemical Sciences, IISER Kolkata. (Dec. 2015).
- Abhisek Ghosal and Amlan K Roy, *DFT of atoms and molecules in Cartesian grid*, on Asian academic seminar and school organized by IISER Kolkata, IACS Kolkata, NINS and IMS of Japan. (6-10 March, 2015).

- **School and Workshop**

- Scikit-Learn Workshop, Northwestern University IT, USA. (10-11 July, 2025)
- Intermediate Python Workshop, Northwestern University IT, USA. (7-9 July, 2025)
- Python Fundamentals, Northwestern University IT, USA. (30th June - 3rd July, 2025)
- 2019 TDDFT School & Workshop: Excited states and dynamics, Rutgers University-Newark, USA. (5-14 August, 2019)
- Computational electronic structure methods for atomic, molecular and solid-state systems, GIAN course at IIT-Bhubaneswar, India. (8-18 July, 2016)

Software Development

FHI-aims Contributions

All-electron electronic structure theory with numeric atom-centered orbitals (2026– Continue).

InDFT (Lead Developer)

A DFT program for atoms and molecules in CCG (Theoretical Chemistry Laboratory, IISER Kolkata, India, 2019) (2014– Continue).

TURBOMOLE Contributions

Program Package for Electronic Structure Calculations (2021-2023).

Publications

Published works

1. Raj Roy and **Abhisek Ghosal***, Simple Model for Challenging Excited States: Low-Lying, Core-Level and Conical-Intersection Regimes, *J. Chem. Theory and Comput.*, accepted, (2026).
<https://doi.org/10.1021/acs.jctc.6c00850>.
2. **Abhisek Ghosal** and George C. Schatz, Autoionized excited states of N₂ molecule using complex-basis function spin-flip coupled cluster theory, *J. Chem. Phys.*, **164**, 014118-1 (2026).
3. Ritaj Tyagi, **Abhisek Ghosal** and Vamsee K. Voora, Optical Absorption spectra of F-Center Defects in LiF using single-pole exchange-correlation and periodic electrostatic embedding, *Phys. Chem. Chem. Phys.*, **27**, 18567-18576 (2025).
4. Nikhil S Chellam, Subhajyoti Chaudhuri, **Abhisek Ghosal**, Sajal K Giri and George C. Schatz, Density Functional Tight-Binding Enables Tractable Studies of Quantum Plasmonics, *J. Phys. Chem. C.*, **129**, 13839-13859 (2025).
5. **Abhisek Ghosal**, Pulkit Joshi and Vamsee K Voora, Taming Negative Ion Resonances Using Nonlocal Exchange-Correlation Functionals, *J. Phys. Chem. Lett.*, **15**, 5994-6001, (2024).
6. **Abhisek Ghosal** and Amlan K. Roy, A real-time TDDFT scheme for strong-field interaction in Cartesian coordinate grid, *Chem. Phys. Lett.*, **796**, 139562, (2022).
7. Raj Roy, **Abhisek Ghosal** and Amlan K. Roy, Charge-Transfer Excitation within a Hybrid-(G) KS Framework through Cartesian Grid DFT, *J. Phys. Chem. A*, **126**, 1448, (2022).
8. **Abhisek Ghosal** and Amlan K. Roy, A self-consistent systematic optimisation of range-separated hybrid functionals from *first principles*, *Mol. Phys.* **e1983056** (2021).
9. Raj Roy, **Abhisek Ghosal** and Amlan K Roy, A Simple Effective SCF Method for Computing Optical Gaps in Organic Chromophores, *Chemistry–An Asian Journal* **16** 2729 (2021).
10. **Abhisek Ghosal**, Tarun Gupta, Kishalay Mahato and Amlan K Roy, The effectiveness of pseudopotential Cartesian-grid KS-DFT on single excitation energies through Becke’s exciton model, *Theor. Chem. Acc.* **140** 1 (2021).

11. Jan Gerit Brandenburg, Kieron Burke, Jannis Erhard, Emmanuel Fromager, **Abhisek Ghosal** and *et al.*, New density-functional approximations and beyond: general discussion, *Faraday Discussions* **224** 166 (2020).
12. **Abhisek Ghosal**, Tanmay Mandal and Amlan K. Roy, Efficient HF exchange evaluation through Fourier convolution in Cartesian grid for orbital-dependent density functionals, *J. Chem. Phys.* **150** 064104-9 (2019).
13. Amlan K. Roy, **Abhisek Ghosal** and Tanmay Mondal, InDFT: A DFT Program for Atoms and Molecules in CCG, IISER Kolkata, India: Theoretical Chemistry Laboratory, (2019).
14. Tanmay Mandal, **Abhisek Ghosal** and Amlan K. Roy, Static polarizability and hyperpolarizability in atoms and molecules through a Cartesian-grid DFT, *Theor. Chem. Acc.* **138** 10 (2019).
15. **Abhisek Ghosal**, Tanmay Mandal and Amlan K. Roy, Density functional electric response properties of molecules in Cartesian grid, *Int. J. Quant. Chem.* **118** e25708 (2018).
16. **Abhisek Ghosal**, Neetik Mukherjee and Amlan K. Roy, Information entropic measures of a quantum harmonic oscillator in symmetric and asymmetric confinement within an impenetrable box, *Ann. Phys. (Berlin)* **528** 796-818 (2016).

Book Chapters

1. **Abhisek Ghosal** and Amlan K. Roy, DFT calculations of atoms and molecules in Cartesian grid”, in *Specialist Periodical Reports: Chemical Modelling, Applications and Theory*, **13** 221-260, Eds. M. Springborg and J.-O. Joswig, Royal Society of Chemistry, London (2016).

Submitted

1. Ken Miyazaki* and **Abhisek Ghosal***, Vibronic anatomy of reverse intersystem crossing through singularity-free second order nonradiative rate formulation, *Review in J. Chem. Theory and Comput.* (2026).
2. **Abhisek Ghosal**, Julio C. V. Chagas and George C. Schatz, A complex absorbing potential based MR-CISD approach to autoionizing 2p-2h resonances: application to N₂, *Submitted in J. Phys. Chem. Lett.* (2026).

Works in Progress

1. **Abhisek Ghosal***, Raj Roy and George C. Schatz*, Probing Resonant Inelastic X-ray Spectrum through a modified Δ SCF-based exciton model, *Manuscript under preparation*.

References

- George C. Schatz
Morrison Professor of Chemistry; Professor of Chemical and Biological Engineering
Department of Chemistry
Northwestern University, 2145 Sheridan Rd., Evanston IL 60208-3113 USA

(847)491-5657 (voice) (847) 491-7713 (fax) email: g-schatz@northwestern.edu
Office location: Room 4017, Ryan Hall, 2190 Campus Dr., Northwestern University

- Prof. Amlan K. Roy
Associate prof., Department of Chemical sciences
IISER Kolkata, Mohanpur-741246, WB, India
E-mail: akroy@iiserkol.ac.in, akroy6k@gmail.com
Webpage: <https://www.iiserkol.ac.in/theochem/index.html>
- Dr. Vamsee K. Voora
Department of Chemistry
TIFR Mumbai, 400005, Maharashtra
Phone: +91 22 2278 2978
E-mail: vamsee.voora@tifr.res.in
Webpage : <https://vkvoora.github.io>
- Dr. Attila Cangi
Head of Department
Machine Learning for Materials Design
Helmholtz-Zentrum Dresden-Rossendorf
- Dr. Niranjana Govind
Environmental Molecular Sciences Laboratory
Pacific Northwest National Laboratory
E-mail: niri.govind@gmail.com
- Prof. Neetu Goel
Department of Chemistry and Center of Advanced Studies in Chemistry
Panjab University, Chandigarh-160014, India
Phone: +91 9815970112
E-mail: neetugoel@pu.ac.in